

Inter-City Freight Load Matching Marketplace

Problem Statement #28 — Requirements Engineering & UML Use-Case Modelling

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a verified Shipper to post a freight shipment by entering cargo details, pickup location, delivery location, required vehicle type, pickup deadline, and auction timeframe.	High	Pass: Valid shipment is posted and visible to eligible carriers. Fail: Rejected if mandatory information is missing or shipper is not verified.	Allows shippers to advertise available freight loads and initiate matching.
FR-002	Functional	The system shall allow verified Freight Carriers with valid permits to view eligible freight shipments and submit competitive bids within the specified auction timeframe.	High	Pass: Valid bid is recorded and shipper notified. Fail: Unverified carrier or carrier with invalid permits cannot bid.	Enables competitive matching while ensuring only qualified carriers participate.
FR-003	Functional	The system shall allow the Shipper to review submitted bids and select a winning Freight Carrier before the auction deadline.	High	Pass: Selected carrier is notified and shipment becomes awarded. Fail: Selection is rejected after auction closes.	Gives the shipper control over carrier selection.
FR-004	Functional	The system shall create a shipment transaction and track predefined delivery milestones between the Shipper and selected Freight Carrier.	High	Pass: Pickup, transit, delivery, and proof-of-delivery milestones are recorded with timestamps. Fail: Unauthorized users cannot modify milestone records.	Provides shipment visibility and a reliable transaction history.
FR-005	Functional	The system shall automatically release the milestone payment held in escrow after the Freight Carrier submits valid electronic proof of delivery and delivery is verified.	High	Pass: Payment is released after valid proof is verified. Fail: Payment remains in escrow when verification fails.	Provides secure automated settlement and prevents premature payment.
NFR-001	Performance & Security	Milestone payments held in escrow shall be disbursed automatically within 60 seconds of verified electronic proof-of-delivery submission, while payment and shipment data shall be protected against unauthorized access.	High	Pass: Testing confirms release within 60 seconds under simulated peak load and prevents unauthorized payment access. Fail: Target latency/security is not met.	Fast and secure settlement is essential for a reliable B2B marketplace.
NFR-002	Availability & Reliability	The system shall maintain reliable availability during active shipment auctions and preserve accepted bids, shipment records, milestone information, and payment transactions without data loss.	High	Pass: Availability/recovery tests show active transactions remain accessible and data is preserved after simulated failures. Fail: Data is lost or unavailable.	Auctions and transactions are time-sensitive; downtime/data loss can cause operational and financial loss.